

✅ DAY 19 — VECTOR MEMORY (FIXED VERSION — EXACT CHANGES)

✅ CURRENT STATUS

Day 1 → Day 18 ✅

✅ WHERE YOU ARE

Only short memory (ChatMemory)

NO long-term semantic memory

✅ WHAT YOU ADD

Add vector memory (NO breaking changes)

✅ STEP 1 — CREATE NEW FILE

NEW FILE

memory/vector_memory.py

✅ ADD THIS CODE (NEW FILE)

```
from embeddings.embedding_service import EmbeddingService
```

```
from embeddings.vector_store import VectorStore
```

```
class VectorMemory:
```

```
    def init(self):
```

```
        self.embedder = EmbeddingService()
```

```
        self.store = VectorStore(collection_name="memory")
```

```
    def add(self, text):
```

```
        embedding = self.embedder.encode(text)[0]
```

```
        self.store.add(
```

```
            ids=[str(hash(text))],
```

```
            documents=[text],
```

```
        embeddings=[embedding],
        metadatas=[{"type": "memory"}]
    )

def search(self, query):

    query_embedding = self.embedder.encode(query)[0]

    results = self.store.query(query_embedding)

    return results["documents"][0]
```

✅ **STEP 2 — ADD IMPORT IN AGENT:**  app/agent.py

✅ **FIND THIS LINE:** from memory.chat_memory import ChatMemory

✅ **ADD BELOW IT :** from memory.vector_memory import VectorMemory

✅ **STEP 3 — MODIFY init:**  app/agent.py

✅ **FIND THIS BLOCK**

```
self.planner = Planner()

self.memory = ChatMemory()
```

✅ **REPLACE WITH**

```
self.planner = Planner()

self.memory = ChatMemory()

self.vector_memory = VectorMemory()
```

✅ **STEP 4 — ADD MEMORY SEARCH (IMPORTANT)**

 app/agent.py

✓ FIND THIS LINE

```
plan = self.planner.create_plan(enriched_query)
```

✓ BEFORE THIS LINE ADD

```
# ✓ VECTOR MEMORY SEARCH
```

```
memory_results = self.vector_memory.search(query)
```

```
memory_text = "\n".join(memory_results[:3])
```

✓ NOW FIND THIS BLOCK

```
enriched_query = f"""
```

```
Conversation so far:
```

```
{past_context}
```

```
Current question:
```

```
{query}"""
```

✓ REPLACE WITH THIS

```
enriched_query = f"""
```

```
Conversation so far:
```

```
{past_context}
```

```
Relevant past memory:
```

```
{memory_text}
```

```
Current question:
```

```
{query}"""
```

✓ STEP 5 — STORE MEMORY: app/agent.py

✓ FIND THIS LINE (VERY END):

```
self.memory.add(query, final_answer)
```

✅ REPLACE WITH

```
self.memory.add(query, final_answer)
```

✅ VECTOR MEMORY STORE

```
self.vector_memory.add(f"User: {query} | Answer: {final_answer}")
```

✅ FINAL FLOW AFTER CHANGE

User Query

↓

Vector Memory ✅

↓

Planner

↓

Tools

↓

Answer

↓

Stored in memory ✅

✅ TEST

Run: `uvicorn app.api:app --reload`

✅ Test:

>> what is ai

>> explain again

>> what did I ask?

✅ EXPECTED

Remembers past context ✅

Better follow-ups ✅

✓ **WHAT YOU BUILT**

Long-term semantic memory

✓ **IMPORTANT**

No existing logic broken ✓

Only **added memory layer**

✓ **NEXT**

DAY 20 → Streaming responses

✓ From next day also I will follow this exact **before → after format**
(so you won't get confused again)